Manindra de Mel

M: +61 498 842 763 | E: manindrademel@yahoo.com.au | LinkedIn | GitHub | manindra.com.au

Education

Australian National University **Jan 2022 – Jun 2026**

Bachelor of Advanced Computing | Major in Machine Learning, Minor in Mathematics

Coursework: functional programming, logic & proofs, array programming, systems, machine learning. ANU Extension — Astrophysics (2021).

Work Experience

Rocket Remit (Lead Software Engineer) **Aug 2023 – present**

- Lead engineer on a cross-border money transfer platform covering 30+ corridors in Asia, Africa, and the Pacific, responsible for mobile, web, backend, data, and infrastructure.
- Built customer-facing apps in Flutter (mobile) and Angular (web), and backend services in Groovy and Spring (Java).
- Own the AWS cloud infrastructure, DevOps, and CI/CD pipelines for build, release, and deployment, and manage the production SQL databases.
- Built financial reconciliation of transfers against partner and bank settlements, plus analytics and business dashboards used by management.
- Designed an ML transaction-monitoring system flagging suspicious activity across remittance flows — integrated into compliance review and reduced manual triage volume by ~40%.
- Analysed 100k+ historical transactions to surface corridor-specific risk patterns; outputs used to tune compliance rules.

Plan4Me (Co-founder / Lead Engineer) **Feb 2026 – present**

- AI travel planner on iOS and Android (plan4me.ai): day-by-day itineraries built from pace, budget, and interests, with live weather, routing, expense tracking, and shareable trips; subscriptions through the App Store and Google Play.
- Flutter client; FastAPI and LangGraph backend on Fly.io with Postgres / PostGIS (Supabase), Redis, and Qdrant; generation progress streams live to the app.
- Cut Google Places spend per trip generation from ~$0.85 to $0.04–0.10 with a local-first place corpus (8,000+ places across 211 cities).
- Scored evaluation suite that gates every change to the generation pipeline.

Stomble (Backend Engineer) **Nov 2022 – May 2023**

- Led a backend team of 4; designed services connecting native client apps to AWS-hosted infrastructure across 10+ microservices.
- Owned AWS configuration, deployment, and operations; raised unit-test coverage to 80%+ and onboarded 5+ engineers.

Independent Software Contractor (manindra.com.au) **2021 – present**

- Full-stack web, data pipelines, scrapers, and AI integrations for clients including Canberra Cloud Kitchen, Our Tutor, Quintet Automotive, and mHits AU.

Select Projects

PlantCLEF 2026 — Co-author, accepted to CLEF 2026 [LINK]

- ANU submission to PlantCLEF 2026 (Raj, de Mel, Wasif, Brake); multi-label species ID across 7,806 long-tailed classes at vegetation-quadrat scale.
- Partial fine-tuning of BioCLIP 2.5 ViT-H/14: only the last 4 transformer blocks unfrozen (lower 28 frozen to preserve the Tree-of-Life prior) with auxiliary cross-entropy heads at genus and family levels.
- Trained on a 2.65M-image enlarged manifest (PlantCLEF 2024 + research-grade iNaturalist pull); inference uses a 4×4 tile grid at 224 and 336 px, class-prior logit adjustment, and an adaptive probability threshold.
- Best public Macro F1 of **0.41826** (private 0.40283); paper appears in CLEF 2026 proceedings, Jena, September 2026.

QEM Mesh Simplification with Skeletonization & CNNs [LINK]

- Structure-aware mesh decimation preserving fine geometric features (2024).

Real-Time Engine Tuning via ML [LINK]

- Live optimisation of internal-combustion engine parameters for efficiency and emissions (2024).

OpenBMB · ChatDev [LINK]

- Multi-agent LLM software-development framework (26k+ stars); contributed deployment and CI tooling to the open release (2023).

PPO Reinforcement Learning for Equities Trading

- Resource-efficient policy-gradient agent; productised as a trading-bot service (2023).

Achievements

- PlantCLEF 2026 — paper accepted to CLEF 2026 proceedings, Jena (co-author, 2026)
- IMC Global Trading Challenge (2025)
- Jane Street Electronic Trading Challenge (2024)
- AI Hack Melbourne — 4th place (2023)
- Australian National Python Convention (PyCon AU) — Speaker (2021)
- Young ICT Explorers — Regional Winner (2021)
- Code Quest, Lockheed Martin — Runners-up (2021)
- Australian Informatics Olympiad — Credit (2019–2021)

Relevant Skills

- Python, Java, Groovy, Dart, TypeScript, Rust, C / C++, Haskell, APL, Assembly
- Machine learning, deep learning, computer vision (PyTorch)
- Reinforcement learning (PPO, policy gradients)
- Foundation model fine-tuning (ViTs), LLM agents (LangGraph)
- Mobile and web: Flutter, Angular, React, Svelte, Vue, React Native, Node
- Backend: Spring, FastAPI, Firebase
- Cloud and DevOps: AWS, GCP, Fly.io, Docker, Linux, CI/CD
- Databases: SQL (Postgres), NoSQL, Firestore
- Functional, procedural, logic, OO paradigms
- Web scraping, data pipelines, ETL

Relevant Interests

Building computers; programming software, websites, and machine learning models; researching mathematical concepts related to computer science; mechanics & working on my vehicle.